

The number of hosts given in the topology only includes PCs, printers, and servers. Sheldon and Leonard are both on **R1-LAN**. Their IP addresses are **192.0.50.69** and **192.127.255.253**, respectively. Penny, who lives in the same building, is also on this network; her IP address is the **second-to-last usable address** of R1-LAN.

a. Penny decides to send an email to all devices on LAN1. **What are the source and destination networks?**

b. Sheldon now plans to expand his office after winning the Nobel Peace Prize and hence designs the topology given above. Apply **VLSM** using the network address **69.0.0.0/15** to efficiently create the subnet addresses for the expanded topology.

c. Leonard says a total of **M** addresses are wasted after applying VLSM. Sheldon replies that **N** addresses would have been wasted if **FLSM** were used. Is $N - M = N / MX(N - 2M)$ correct?

[You do not need to show FLSM steps]

